E7: Introduction to Computer Programming
Final Exam – Multiple Choice
Version A1

Thursday, December 14

Full Name

Student ID Number

Instructions:

- This includes the questions of Part I (Multiple Choice). Part II is in a separate exam booklet.
- Write your name above AND on the bubble sheet.
- Mark the correct version letter AND version number on the bubble sheet.
- This exam is DOUBLE-SIDED. Make sure to read both sides of each page.
- This part of the exam is worth a total of 100 points.
- There are a total of 50 questions.
- Each question is worth 3 points, for a total possible of 150 points.
- There is **ONE and only one** correct choice for each question.
- Mark your answers on the bubble sheet with pen or pencil.
- Multiple bubbles, incomplete bubbles, or stray marks will be marked incorrect.
- Anything written on this exam will NOT be graded.
- You may use this booklet as scratch.
- ONLY answers on the bubble sheet will be graded.
- At the end of the exam, hand in the completed bubble sheet AND the exam.

Good luck!

1. The central difference method is known as 'second order' because

   A. it takes more time to run
   B. it results in high error
   C. if we divide the step size by 2, the error decreases by a factor of 2
   D. the derivative is computed by dividing the difference by twice the step size
   E. if we divide the step size by 2, the error decreases by a factor of 4

2. After running the following Python code, what will be the final value of `pos`?

```
import numpy as np
pos = np.arange(-4, 4)
pos[pos >= 2] = pos[pos >= 2] - 2
```

   A. `[-4 -3 -2 -1 0 1 2 1 2]`
   B. None of the options
   C. `[-4 -3 -2 -1 0 1 0 1 2]`
   D. `[-4 -3 -2 -1 0 1 2 1]`
   E. `[-4 -3 -2 -1 0 1 0 1]`

3. Assume that the MATLAB code below has been executed.

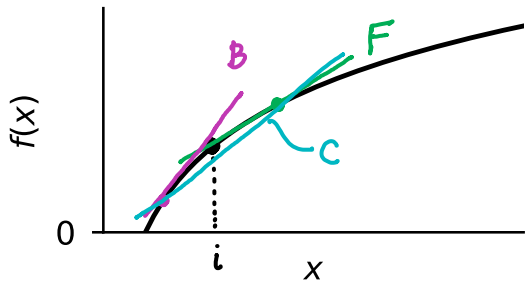
```
A = [1, 0; 2, 3; 5, 4];
B = [1, 0, 0; 0, 1, 0];
```

   Which of the following MATLAB expressions will NOT result in an error?

   A. `A .* B`
   B. `A - B`
   C. None of the options result in an error
   D. `A + B`
   E. `A * B`

4. For the function $f(x)$ shown below, which differentiation method will likely yield the highest (i.e., largest) numerical estimate of the derivative of the function at any point?

   A. Forward difference
   B. Backward difference
   C. Central difference
   D. Midpoint rule
   E. Simpson's rule



5. You need to perform a numerical computation, which uses a step size h. You've come across various methods that exhibit distinct orders of error. Which method would you select?

   A. Method with $O(h)$
   B. Method with $O(h^4)$
   C. Method with $O(h^3)$
   D. Method with $O(h^2)$
   E. Method with $O(\log h)$

6. After running the following Python code, what will be the final value of `var`?

```
var = 2
var += 2 ** 4 / 2
```

   A. 4
   B. 10
   C. 8
   D. 2
   E. 6

7. Let `y` and `t` be two NumPy arrays of the same size. Which numerical differentiation method does the following function correspond to?

```
def numerical_differentiation(y, t):
    return (y[1:] - y[:-1]) / (t[1:] - t[:-1]), t[1:]
```

   A. Central difference
   B. Forward difference
   C. Simpson's rule
   D. Backward difference
   E. Midpoint rule

8. While using ChatGPT to better understand root finding, it mentioned the following sentence, *"Rather than aiming for precisely zero error, numerical algorithms employ tolerance thresholds to decide when to terminate the iterations."* This statement is

   A. true because numerical root finding methods were developed more than 200 years ago
   B. false
   C. true because the algorithm must account for truncation errors
   D. true because the algorithm must account for round-off errors
   E. true because ChatGPT is always correct

9. Which of the following is True regarding numerical differential equation solvers?

   A. They are not suitable for systems of differential equations
   B. They cannot be used for solving higher order differential equations
   C. Increasing the step size increases accuracy
   D. They are iterative in nature
   E. Increasing the step size increases computation time

10. Round-off errors occur because numbers in computers are

   A. represented using a fixed number of bits
   B. represented using the binary system
   C. represented using the decimal system
   D. truncated from an infinite to a finite sum
   E. A and B

11. What will the expression `int(5.8) + float(4.5)` evaluate to in Python?

   A. 10.3
   B. 9.5
   C. 10.5
   D. 10
   E. 9

12. The decay of a radioactive isotope, can be modeled by the following differential equation:

$$\frac{d^2N}{dt^2} + 100\frac{dN}{dt} + N = 0$$

where N is the concentration of the isotope at a given time. Let the state be: $S = \begin{bmatrix} N \\ N' \end{bmatrix}$

Which of the following is Forward Euler's solution with a step size of 0.1 in matrix form?

   A. $\begin{bmatrix} N_{i+1} \\ N'_{i+1} \end{bmatrix} = \begin{bmatrix} 0 & 0.1 \\ -0.1 & -10 \end{bmatrix} \begin{bmatrix} N_i \\ N'_i \end{bmatrix}$
   B. $\begin{bmatrix} N_{i+1} \\ N'_{i+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & -99 \end{bmatrix} \begin{bmatrix} N_i \\ N'_i \end{bmatrix}$
   C. $\begin{bmatrix} N_{i+1} \\ N'_{i+1} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & -100 \end{bmatrix} \begin{bmatrix} N_i \\ N'_i \end{bmatrix}$
   D. $\begin{bmatrix} N_{i+1} \\ N'_{i+1} \end{bmatrix} = \begin{bmatrix} 1 & 0.1 \\ -0.1 & -9 \end{bmatrix} \begin{bmatrix} N_i \\ N'_i \end{bmatrix}$
   E. $\begin{bmatrix} N_{i+1} \\ N'_{i+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} N_i \\ N'_i \end{bmatrix}$

13. You want to compute $I = \int_0^5 f(x) = xe^{x^2}$ using `scipy.integrate.quad(func, a, b, ...)`, where `func` is a function object of the function you want to integrate and `a, b` are two scalar values representing the lower and upper limits of integration, respectively. The function also takes additional optional arguments. What is wrong, if anything, with the code below?

```
import scipy
import numpy as np
def func(x):
    print(x * np.exp(x ** 2))
I = scipy.integrate.quad(func, 0, 5)
```

A. `func` should have `return` instead of `print()`

B. Nothing is wrong with the code (it will correctly compute I)

C. `scipy.integrate.quad` should have an additional argument `args=(...)`

D. The code has incorrect indentation

E. The mathematical expression `x * np.exp(x ** 2)` has wrong order of precedence

14. The Bisection method is a numerical method that is used to

A. integrate a function

B. differentiate a function

C. find the maximum value of a function

D. optimize functions

E. find a root of a function

15. Programming makes your work

A. transparent

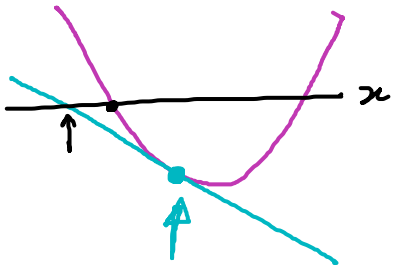
B. reproducible

C. All of the options

D. shareable

E. efficient

16. Which of the following is False regarding the Newton-Raphson method?



A. It requires knowing or estimating the derivative of the function

B. It will converge to the root closest to the initial guess

C. It is used to estimate roots of a function

D. It may converge to a different root than the desired one

E. In general, it converges to a solution faster than the Bisection method

17. Regression is used to find

A. data points that are as close as possible to the actual data

B. a curve that best fits the data by minimizing some measure of error

C. a curve that best fits the data by maximizing some measure of error

D. the values at which a function is equal to zero

E. a curve that exactly fits the data

18. Consider the following Python code, where `a` is a data structure.

```
s = 0
for i in range(len(a)):
    if a[i] > 2:
        s += 1
```

Which of the following options is equivalent to the above code? Note that `sum` is a built-in function that returns the sum of the elements in a data structure.

A.
```
s = sum(a > 2)
```

B.
```
s = sum([2 for i in a if i > 1])
```

C.
```
s = lambda a: sum(a > 2)
```

D.
```
s = 0
for i in a:
    if a[i] > 2:
        s += 1
```

E.
```
s = 0
for i in range(len(a)):
    if a[i] > 2:
        s + 1
```

19. Let `f` be a function object you want to integrate and `a, b, h` be three scalar values representing the lower integration bound, upper integration bound, and step size, respectively. Which numerical integration method does the following function correspond to?

```
def numerical_integration(f, a, b, h):
    x = np.arange(a, b, h)
    return np.sum(h * f(x + h/2))
```

A. Left or Right Riemann (either one)

B. Midpoint rule

C. None of the methods

D. Trapezoid rule

E. Simpson's rule

$\sum h f\left(x+\frac{h}{2}\right)$

Midpoint

20. Which of the following analogies was given in the lecture when introducing functions?

A. Functions are like televisions

B. Functions are like cars

C. Functions are like pizza

D. Function are like toasters

E. Functions are like refrigerators

21. Your friend wrote the root finding function below, which takes as input a function object `f`, two scalar values `a, b` representing an initial range, and a scalar value `tol` representing the tolerance. The absolute value of the function at the acceptable root should be less than `tol`. What is wrong, if anything, with the function? Note that line numbers are shown to the left.

```
1    import numpy as np
2    def roots(f, a, b, tol):
3        assert f(a) * f(b) < 0, 'Invalid [a, b]'
4        midpoint = (a + b) / 2
5        while np.abs(f(midpoint)) >= tol:
6            if f(midpoint) * f(a) >= 0:
7                a = midpoint
8            else:
9                b = midpoint
10       return midpoint
```

Bisection

f(m) & f(a)

a=m

b=m

midpoint = (a+b)/2

A. Nothing is wrong with the function

B. The logical expression on line 5 is incorrect

C. The assert statement on line 3 is incorrect

D. There is a missing line of code that should be added after line 9

E. The assignment expressions on lines 7 and 9 should be swapped

22. Consider the following MATLAB code:

```
midterm_1 = [95.5, 49, 89.5, 71.5, 43];
midterm_2 = [86.5, 78, 90.5, 71.5, 78];
compare = (midterm_1 < midterm_2) & (midterm_1 < 50);
```

Which option yields an output equivalent to that of `compare` in the code above?

A. `compare = (midterm_1 < midterm_2) .* (midterm_1 < 50);`

B. `compare = (midterm_1 < midterm_2) + (midterm_1 < 50);`

C. `compare = (midterm_1 < midterm_2) and (midterm_1 < 50);`

D. `compare = (midterm_1 < midterm_2) * (midterm_1 < 50);`

E. `compare = (midterm_1 < midterm_2) && (midterm_1 < 50);`

23. Your friend, who hasn't taken E7, is trying to solve a system of 5 linear equations with 4 unknowns: x_1, x_2, x_3, x_4. After overhearing you while watching the E7 recordings, they believe they know how to express this system in matrix form and came up with the form shown below. How can you fix the matrix form of this system of linear equations?

$$\underbrace{\begin{bmatrix} 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 100 \\ 10 \\ 90 \\ 70 \\ 50 \end{bmatrix}}_{\mathbf{b}}$$

A. Add one more column to matrix **A**

B. Add one more row to matrix **A**

C. Remove one row from vector **b**

D. None of the options (i.e., the system is correctly defined)

E. Add one more row to vector **x**

24. The accuracy of a truncated Taylor series approximation can be improved by

A. using more subintervals

B. using a smaller step size

C. using more terms

D. A and B

E. A, B, and C

25. What will be displayed after running the following Python code?

```
def fun(a, b):
    if a < b:
        return a - b
    else:
        return fun(a-2, b)

print(fun(5, 2))
```

A. Error

B. 1

C. -1

D. 3

E. None of the options

26. You defined a system of linear equations in MATLAB and obtained the following results.

```
>> A = [2, 1, 2; 1, -2, -1; 4, 0, -1];
>> b = [6; 5; 8];
>> rank_A = rank(A)
rank_A =
       3

>> rank_Ab = rank([A, b])
rank_Ab =
       3
```

If the matrices are correctly defined, this system of linear equations has

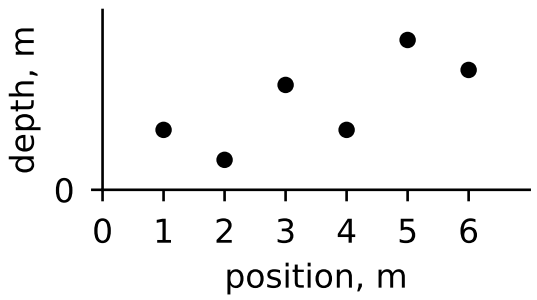
A. It is not possible to determine anything about the solution based on the given information
B. no solution
C. a solution but it is not possible to determine whether it is unique or not based on the given information
D. a unique solution
E. infinite solutions

27. You collected a data set of temperature measurements taken every 15 minutes. You are interested in analyzing this data, which might require unit conversion and other numerical computations. What is the recommended data structure for storing this data set in Python?

A. Range
B. List
C. NumPy Array
D. Tuple
E. String

28. You've collected a data set containing the floor depth of a lake at various positions along a specific cross-section, as shown in the figure below. To numerically integrate this data with a step size of 1 m, which, if any, of the following methods cannot be used?

A. None of the options (i.e., all can be used)
B. Simpson's rule
C. Midpoint rule and Simpson's rule
D. Midpoint rule
E. Left and Right Riemann

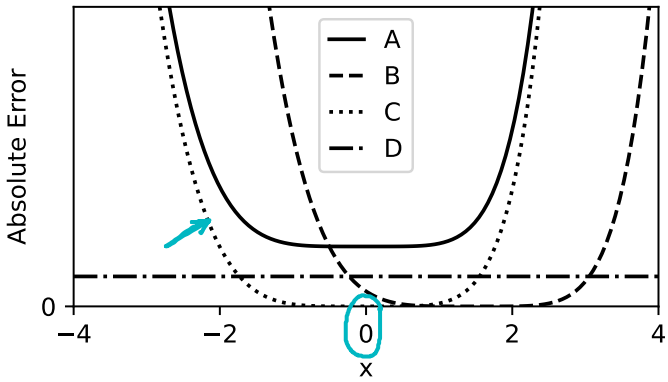


29. What is the highest number that can be represented with a 3-bit signed binary representation?

A. 4
B. 3
C. 2
D. 7
E. 14

30. After computing the Taylor series expansion for $f(x) = e^x$ around the point $a = 0$, you plotted the absolute value of the error as a function of x. Which of the curves in the figure below represents the correct trend of the absolute error?

A. A
B. B
C. C
D. D
E. None of the options



31. How can the error of numerical integration methods be reduced, and what is the resulting consequence?

A. Decrease step size, increase computation time
B. It depends on the integration method
C. Increase step size, decrease computation time
D. Decrease step size, decrease computation time
E. Increase step size, increase computation time

32. The Van der Pol equation describes the motion of dynamic systems as:

$$\frac{d^2x}{dt^2} - \mu\left(1 - x^2\right)\frac{dx}{dt} + x = 0$$

where μ is some constant. What can you say about the Van der Pol equation?

A. None of the options
B. It is a second order ordinary differential equation
C. It is a third order ordinary differential equation
D. It is a third order partial differential equation
E. It is a second order partial differential equation

33. What will be displayed after running the following Python code?

```
import numpy as np
midterm_1 = np.array([95.5, 49, 89.5, 71.5, 43])
midterm_2 = np.array([86.5, 78, 90.5, 71.5, 78])
print(midterm_1 < midterm_2)
```

A. `[False True True True True]`

B. `[False True True False True]`

C. `False`

D. None of the options

E. `[True False False True False]`

34. For the function $f(x)$ shown below, which integration method will likely yield the highest (i.e., largest) numerical estimate of the area under the curve?

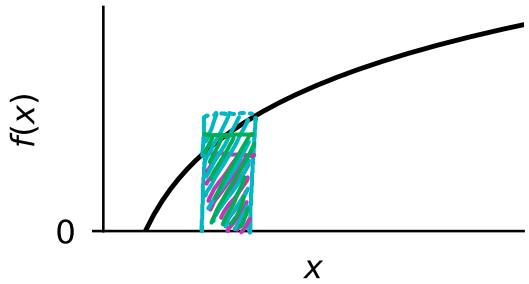
A. Left Riemann

B. Midpoint rule

C. Simpson's rule

D. Right Riemann

E. Trapezoid rule



35. Assume that the following MATLAB function, `fun()`, has been defined.

```
function [c] = fun(a, b)
c = 0;
if a > b
c = a - b;
end
if c < 10
return
else
c = 10;
end
end
```

What will be the output of `fun(5, 2)`?

A. 0

B. 10

C. 3

D. None of the options

E. -3

36. Assume `wind` is a list that includes wind speeds. Consider the following Python code.

```
m = len(wind)
for i in range(m):
    wind[i] *= 0.3048
```

Which of the following options is equivalent to the above code?

A.
```
for value in wind:
    value *= 0.3048
```

B.
```
m = len(wind)
for i in len(m):
    wind[i] *= 0.3048
```

C.
```
m = len(wind)
for i in m:
    wind[i] *= 0.3048
```

D.
```
for j, value in enumerate(wind):
    wind[j] = value * 0.3048
```

E.
```
m = len(wind)
for i in range(m):
    wind[i] = 0.3048
```

37. What will be displayed after running the following Python code?

```
def physics(mass, net_force):
    if mass > 1:
        force = 10 * mass
        friction = net_force - force
        if friction > 0:
            return friction
        else:
            return 'Negative friction detected'
    else:
        return 0

print(physics(2, 10))
```

A. Error

B. `'Negative friction detected'`

C. `0`

D. `-10`

E. `10`

38. Which of the following is NOT true regarding differences between Python and MATLAB?

    A. `[9, 1, 5]` is a `list` in Python but a `double array` in MATLAB

    B. Python uses 0-based indexing whereas MATLAB uses 1-based indexing

    C. Many mathematical functions are built into MATLAB but need to be imported in Python

    D. Indentation is required in Python but optional in MATLAB

    E. None of the options (i.e., all are True)

39. What will be displayed after running the following Python code?

```
def fun(a, b):
    if a > b:
        c = a - b
    else:
        c = b - a
    return c
fun(5, 2)
print(c)
```

    A. 3

    B. -2

    C. 5

    D. 2

    E. Error

40. After running the following Python code, what will be the final value of `f`?

```
m = 5
a = 2
if a == 2:
    f = m * a
    a = 10
if a == 10:
    m = 10
    f = f / 10
elif m == 10:
    f = 2
```

    A. `f` will not be defined

    B. 1

    C. 5

    D. 10

    E. 2

41. What happens if a recursive function does not have any base case(s)?

    A. The function will have an infinite loop
    B. The function will be iterative
    C. The function will have an infinite recursion
    D. The function will return an incorrect answer
    E. The function will return a correct answer but it will take too long

42. What was the secondary programming language used in E7 this semester?

    A. JavaScript
    B. LOLCODE
    C. Python
    D. MATLAB
    E. C++

43. What was the primary programming language used in E7 this semester?

    A. JavaScript
    B. C++
    C. LOLCODE
    D. Python
    E. MATLAB

44. Which of the following errors is certain to lead a program to crash?

    A. Syntax error
    B. Round-off error
    C. Truncation error
    D. Logic error
    E. Runtime error

45. Which of the following is not a valid number in base 4 representation?

    A. 00000
    B. 10401
    C. 10001
    D. 32221
    E. 12003

46. What will be displayed after running the following Python code?

```
data1 = [0, 1, 1, 0, 1]
data2 = [2, 2]
print(data1 + data2)
```

A. `[0, 1, 1, 2, 3]`
B. Error
C. `[2, 3, 1, 0, 1]`
D. `[0, 1, 1, 2, 2]`
E. `[0, 1, 1, 0, 1, 2, 2]`

47. Order the following integration methods from the most accurate (least error) to least accurate (highest error) for the same step size h:

A. Simpson's, Trapezoid, Left Riemann
B. Trapezoid, Simpson's, Left Riemann
C. Simpson's, Left Riemann, Trapezoid
D. Left Riemann, Trapezoid, Simpson's
E. Trapezoid, Left Riemann, Simpson's

48. Which error among the following options is not caused by incorrectly written code?

A. None of the options (i.e., all are caused by incorrectly written code)
B. Logic error
C. Round-off error
D. Runtime error
E. Syntax error

49. Which of the following can be effectively solved using numerical methods?

A. Finding roots to algebraic equations
B. Finding the integral of a function
C. Finding solutions to differential equations
D. All of the options
E. Finding the derivative of a function

50. What is the preferred method for implementing code that performs the same task on an input?

A. Using iteration and conditional statements
B. Using recursion
C. Using iteration
D. Using functions
E. Using conditional statements